Aaryav Gangrade

+91-9691354445 | aaryav1805.gangrade@gmail.com
linkedin.com/in/aaryav-gangrade | github.com/AaryavGangrade | aaryavgangrade.github.io/portfolio

Education

Vellore Institute of Technology — Chennai, IN
B.Tech in Computer Science and Engineering; CGPA: 7.61/10 — *2022 – 2026*

Mount Carmel School — Bhopal, IN
Class XII (PCM); 90.2% — *2022*

Experience

NZESSENTIALS — Christchurch, NZ (remote)
Freelance Python Developer — *May 2025 - July 2025*

- Built a digital signature platform with web and desktop applications for secure signature collection and verification
- Created REST APIs to connect different parts of the application
- Built a fast system to store, retrieve, and verify digital signatures

Linux Club, VIT Chennai — Chennai, IN
Operations Department Lead — *March 2023 – March 2024*

- Led a team of 150 in organizing workshops and events promoting Linux and open-source software.
- Coordinated logistics for club activities, ensuring smooth operations and effective communication.
- Secured sponsorships totaling Rs 1,00,000 to support club events and initiatives, enhancing resources and outreach.

Projects

Agentic Graph-RAG for Codebase Intelligence | *Python, FAISS*

- Built a RAG system that parses Python codebases into a searchable structure of functions, classes, and call relationships for deep, context-aware code understanding
- Benchmarked on the production `psf/requests` repo (31,441 LOC, 26 files): parsed the full codebase in 356 ms (∼88,400 LOC/sec) and returned hybrid search + graph-expanded results in ∼340 ms, with vector/keyword lookups running in sub-millisecond time
- Compressed 31,441 LOC of raw codebase context down to under 700 relevant lines per query (97.8–98.6% reduction), keeping only the most relevant code in the LLM's prompt

NSE W/M-Curve Pattern Scanner | *Python, yfinance, pandas, ThreadPoolExecutor*

- Built an automated technical-pattern scanner across 2,600+ NSE equities to detect W-curve (double-bottom) and M-curve reversal setups, using parallelized data pipelines, parquet caching, and a daily scheduler generating HTML signal reports
- Backtested W-curve strategy (TP 15% / SL 5% / 20-day hold / min score 90) to +22.70% cumulative return over 2 years across 624 trades, vs. Nifty 50 benchmark's -0.07% over the same period
- Backtested M-curve strategy (TP 10% / SL 5% / 20-day hold / min score 70, Nifty 500 universe) to +11.36% cumulative return, outperforming the Nifty 50 benchmark

Legal.ly | *Kotlin*

- Developed a native Android application during Smart India Hackathon to improve legal information accessibility
- Integrated chatbot-based assistance system to handle user queries and provide relevant legal guidance
- Implemented search and filtering mechanisms to help users find lawyers and legal information efficiently
- Developed structured backend communication for retrieving and displaying legal data dynamically

Technical Skills

Languages: Python, Java, C++, JavaScript, Kotlin, SQL
Frameworks/Tools: FastAPI, Firebase, Streamlit, Git, Pytest
AI/ML: NLP, Embeddings, RAG Systems, LLM Agents
Web: React, REST APIs, JSON
Libraries: Pandas, NumPy, yfinance
Systems: Linux, Windows
Android: Kotlin, Android SDK, Jetpack Components, REST API Integration, Material Design